

Skills Summary

Locating Zeros and Turning-Point Constraints

Use this reference while completing your worksheet or when studying.

Skill 1 — Factoring completely to locate the zeros

Order of attack: common factor first, then a difference of squares, then a quadratic pattern (including a quadratic in x^2), then grouping or a known zero.

Pulling out the common x is what keeps the zero at 0 visible — dividing it away loses a real zero.

Example: Find all real zeros of $p(x) = x^3 - 2x^2 - 8x$.

Step 1. Every term carries an x , so factor that out first; the leftover quadratic is then an ordinary trinomial.

Step 2. $x(x^2 - 2x - 8) = x(x - 4)(x + 2)$.

$$\Rightarrow x = 0, x = 4, x = -2$$

Try it: Find all real zeros of $p(x) = x^3 - x^2 - 12x$.

check: $x = 0, x = 4, x = -3$

Skill 2 — Multiplicity against degree

Rule. A factor $(x - r)^m$ gives a zero at r of multiplicity m , and the multiplicities add up to the degree.

Odd multiplicity means the curve crosses; even multiplicity means it touches and turns back.

Example: Find the zeros of $p(x) = (x - 2)^2(x + 5)$ and check the degree.

Step 1. The factored form already shows both zeros, so read the exponents rather than expanding.

Step 2. Multiplicity 2 at $x = 2$ (a touch) and multiplicity 1 at $x = -5$ (a crossing); $2 + 1 = 3$, the degree.

$$\Rightarrow x = 2, x = -5$$

Try it: Find the zeros of $p(x) = (x + 3)^2(x - 1)$ and say which one is a touch.

check: $x = -3$ touch, $x = 1$ crossing

(!) Watch out: a repeated factor does not always mean a touch. x^3 is a repeated factor with multiplicity 3, which is odd, so the curve crosses there.

Skill 3 — Bounding the turning points

Rule. A polynomial of degree n has at most n real zeros and at most $n - 1$ turning points — its derivative has degree $n - 1$, so it can vanish at most $n - 1$ times.

From below: each gap between two consecutive simple zeros forces at least one turning point.

Example: A polynomial has degree 7. What is the greatest possible number of turning points?

Step 1. The bound comes from the degree alone, so no coefficients are needed — subtract one.

Step 2. $7 - 1 = 6$.

$\Rightarrow$ **6**

Try it: A polynomial has degree 10. Greatest possible number of turning points?

6 :peəp